Uber Trips Analytics

CAPSTONE PROJECT

Presented by
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Trips | Revenue | Geography | Passengers | Demand Forecast



Uber | Trips Analytics

CAPSTONE PROJECT

BETTER INSIGHTS
FOR A SMARTER RIDE

Project Overview

About the Project

This project analyzes Uber green taxi trip data from New York City to uncover key trends, improve operational efficiency, and support data-driven decisions for better fleet management, dynamic pricing, and an enhanced customer experience.

Business Problem

Understanding trip patterns, revenue trends, geographic demand, and passenger behavior is essential to optimize resources, reduce wait times, and maximize profitability.

Goal

Use SQL and Power BI to analyze trip and revenue data, identify key patterns, and build a comprehensive dashboard with actionable insights and a demand forecasting model.

Key Focus Areas

Trip Trends
Volume, frequency and seasonal patterns

Revenue Analysis
Fare trends, revenue breakdowns and profitability insights

Geographic Insights
Popular zones, top routes and borough analysis

Passenger Behavior
Load patterns, payment methods and tip trends

Demand Forecasting
Predict future demand using historical data



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Project Objectives & Business Problem

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Business Problem

The Uber Taxi & Limousine Commission needs to analyze green taxi trip data to identify trends, improve operational efficiency, optimize fleet management, set dynamic pricing, and enhance customer experience in New York City.

Why It Matters?

- Helps in better fleet allocation
- Improves pricing strategy
- Increases revenue and operational efficiency
- Enhances passenger satisfaction

Project Objectives

01 **Understand Trip Volume & Revenue Trends**
Analyze historical trip data to identify patterns, seasonality, and growth trends.

02 **Identify Peak Demand & High-Demand Locations**
Find the busiest days, hours, and zones to improve fleet deployment and reduce wait times.

03 **Analyze Fare, Tips & Payment Patterns**
Understand passenger behavior and payment preferences to optimize pricing and incentives.

04 **Support Fleet Efficiency & Pricing**
Provide data-driven recommendations for fleet scheduling and dynamic pricing strategies.

05 **Build a Predictive Model**
Forecast future demand to ensure better resource planning and service reliability.



Uber | Dataset, Tools & Methodology

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Dataset

NYC Green Taxi Trip Records

- Historical trip data from New York City
- Green Taxi & Limousine Commission
- Data covers multiple years with detailed trip, fare, and location information

Tools Used

SQL
Data extraction, transformation & analysis

Power BI
Visualization & Dashboard Creation

Methodology

01 **Data Collection**
Importing CSV data to SQL

02 **Data Cleaning**
Handling missing values, duplicates, and data type conversion

03 **SQL Analysis**
Solving 10 business questions

04 **Dashboard Development**
Building 5 interactive Power BI dashboards

05 **Insights & Recommendations**
Deriving actionable business insights

Key Fields in the Dataset

- tpep_pickup_datetime (Pickup date & time)
- tpep_dropoff_datetime (Dropoff date & time)
- pickup_borough (Pickup borough)
- dropoff_borough (Dropoff borough)
- pickup_zone (Pickup zone)
- dropoff_zone (Dropoff zone)
- fare_amount (Trip fare)
- tip_amount (Trip amount)
- payment_type (Payment method)
- passenger_count (Number of passengers)
- trip_distance (Distance in miles)



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SQL Analysis — Question 01

YEARLY TRIP TRENDS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL Query

```sql
-- 1. Yearly Trip Trends: Calculate total trips per year and percentage change year-over-year.
WITH yearly_trips AS (
  SELECT 2017 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2017
  UNION ALL
  SELECT 2018 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2018
    UNION ALL
  SELECT 2019 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2019
     UNION ALL
  SELECT 2020 AS trip_year, COUNT(*) AS total_trips FROM stg_uber_2020
),
yearly_comparison AS (
  SELECT trip_year,total_trips,LAG(total_trips) OVER (ORDER BY trip_year) AS previous_year_trips FRO
  FROM yearly_trips
  )SELECT trip_year, total_trips, previous_year_trips,ROUND((total_trips - previous_year_trips)*
100.0 / NULLIF(previous_year_trips, 0),2)
  ) AS yoy_percentage_change
FROM yearly_comparison
ORDER BY trip_year;
```

Query Result: Yearly Trip Trends

Result Grid | Filter Rows: | Export: | Wrap Cell Co

trip_year	total_trips	previous_year_trips	yoy_percentage_change
2017	1048575	NULL	NULL
2018	8807303	1048575	1048575
2019	6045506	8807303	-31.30
2020	1735507	6045506	-71.29



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SQL Analysis — Question 02

MONTHLY REVENUE INSIGHTS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL QUERY

```
#Q2:   Monthly Revenue Insights: Find monthly total revenue and average revenue per trip.
), WITH mth_trip AS (SELECT zapp_lookup_datetime AS pickup_date,
  total_amount AS revenue FROM fact_trip_data
  UNION ALL
  SELECT zapp_lockup_datetime, total_amount
  FROM mth_trip,2018
), ranked AS (
  SELECT zapp_lockup_datetime, total_amount FROM mth_trip, 2018
  UNION ALL
  SELECT zapp_lookup_datetime, total_amount FROM mth_trip, 2019
), SELECT year(pickup_date) as trip_year,MONTH(pickup_date) as trip_month,ROUND(SUM(pickup_date) as trip_month,
  round(AVG(revenue), 2) as total_revenue, round(AVG(revenue), 2) as average_revenue_per_trip,
  FROM ranke fao Year(pickup_date) IS NOT NULL AnD MOTH(pickup_date) GETCIER 2017 ROM 2020 AnD MONTH(pickup_date) co 0
  GROUP BY
  year(pickup_date), MONTH(pickup_date), MONTHNAME(pickup_date)
  ORDER BY trip_year, trip_month;
```

QUERY RESULT

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

trip_year	trip_month	month_name	total_revenue	average_revenue_per_trip
2017	1	January	6216.61	14.69
2017	2	February	4037.88	13.09
2017	3	March	4083.43	13.32
2017	4	April	3459.07	13.35
2017	5	May	3918.83	13.31
2017	6	June	4375.57	13.73
2017	7	July	3689	14.62
2017	8	August	4883	14.1
2017	9	September	3734.25	12.62
2017	10	October	6390.22	13.35
2017	11	November	3966.56	14.11
2017	12	December	4573.08	13.47
2018	1	January	11113.31.16	14
2018	2	February	10907.95.60	14.81
2018	3	March	12810.98.03	14.51



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SQL Analysis — Question 03

DAILY & WEEKLY TRIP PATTERNS

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SQL Query

```sql
-- 3. Daily & Weekly Trip Patterns: Find trips by day of week and hour.
WITH all_trips AS (
   SELECT lpep_pickup_datetime AS pickup_date,
          total_amount AS total_amount
    FROM stg_uber_2017
   UNION ALL
  SELECT lpep_pickup_datetime AS pickup_date,
          total_amount AS total_amount
    FROM stg_uber_2018
   UNION ALL
  SELECT lpep_pickup_datetime AS pickup_date,
          total_amount AS total_amount
    FROM stg_uber_2019
   UNION ALL
  SELECT lpep_pickup_datetime AS pickup_date,
          total_amount AS total_amount
    FROM stg_uber_2020
),
daily_hourly_trips AS (
  SELECT DATE(pickup_date) AS day_num,
         EXTRACT(HOUR FROM pickup_date) AS trip_hour,
         COUNT(*) AS total_trips,
         AVG(total_amount) AS avg_trips
   FROM all_trips
   WHERE pickup_date IS NOT NULL AND YEAR(pickup_date) BETWEEN 2017 AND 2020
   GROUP BY
        YEAR(pickup_date), MONTH(pickup_date), MONTHNAME(pickup_date)
),
ORDER BY trip_year, trip_month;
```

Query Result: Daily & Weekly Trip Patterns

day_of_week	trip_hour	time_slot	avg_trips
Friday	17	17:00 - 18:00	1166.86
Friday	19	19:00 - 20:00	1120.03
Thursday	18	18:00 - 19:00	1115.69
Friday	16	16:00 - 17:00	1086.65
Thursday	17	17:00 - 18:00	1085.39



Uber | SQL Analysis — Question 04

BOROUGH TRIP DISTRIBUTION

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FOR A SMARTER RIDE

SQL Query

```sql
# 4. Borough Trip Distribution: Find the percentage of trips starting in each pickup_borough.
with all_trip as (
select PULocationID from stg_uber_2017
union all
select PULocationID from stg_uber_2018
union all
select PULocationID from stg_uber_2017
union all
select PULocationID from stg_uber_2019
union all
select PULocationID from stg_uber_2020
), Borough_trip as (select z.Borough as pickup_Borough , count(*) as Total_trips from all_trip t join taxi_zones z on t.PULocationID=z.LocationID
group by z.Borough
)SELECT pickup_borough,total_trips, ROUND(
        total_trips * 100.0 / SUM(total_trips) OVER (),
        2
    ) AS trip_percentage
FROM borough_trip
ORDER BY trip_percentage DESC;
```

Activate Windows

Query Result: Borough Trip Distribution

Result Grid | Filter Rows: | Export: | Wrap Cell Con

pickup_borough	total_trips	trip_percentage
Manhattan	6241731	33.40
Brooklyn	5784791	30.96
Queens	5404927	28.93
Bronx	1210186	6.48
Unknown	36326	0.19



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SQL Analysis — Question 05

TOP 10 PICKUP-DROPOFF ROUTES

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL Query

```sql
-- 5. Top PickUp-Dropoff Routes: Identify the top 10 most frequent pickup_zone to
dropoff_zone combinations.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID
     FROM stg_uber_2017
    UNION ALL
    SELECT PULocationID, DOLocationID
     FROM stg_uber_2018
    UNION ALL
    SELECT PULocationID, DOLocationID
     FROM stg_uber_2019
    UNION ALL
    SELECT PULocationID, DOLocationID
     FROM stg_uber_2020
),
borough_trip AS (select  z.Borough as pickup_zone, d.Borough as dropoff_zone,
    COUNT(*) as Total_trips from all_trip t
    join `taxi_zones` pu on t.PULocationID = pu.LocationID
    join `taxi_zones` doff
      on t.DOLocationID = doff.LocationID
    group by
      pu.zone,
      doff.zone
    order by
    total_trips DESC
    LIMIT 10;
```

Query Result: Top 10 Routes

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

pickup_zone	dropoff_zone	total_trips
Astoria	Astoria	212538
East Harlem South	East Harlem North	199548
Central Harlem	Central Harlem North	168487
East Harlem North	East Harlem South	165029
Forest Hills	Forest Hills	146291



Uber | SQL Analysis — Question 06

PASSENGER LOAD PATTERNS

BETTER INSIGHTS
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SQL Query

```sql
-- 6. Passenger Load Patterns: Find the average passenger_count for trips in each borough.
with all_trips as (
  SELECT passenger_count, PULocationID from stg_uber_2017
  UNION ALL
  SELECT passenger_count, PULocationID from stg_uber_2018
  UNION ALL
  SELECT passenger_count, PULocationID from stg_uber_2019
  UNION ALL
  SELECT passenger_count, PULocationID from stg_uber_2020
)

SELECT z.Borough, avg(t.passenger_count) as avg_passengers from all_trips t
join taxi_zones z
on t.PULocationID = z.LocationID
group by Borough;
```

Query Result: Average Passengers per Borough

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

Borough	avg_passengers
Manhattan	1.288655919547183
Brooklyn	1.3282517457731373
Queens	1.4009973813224699
Bronx	1.2909298846621458
Unknown	1.238132887334651



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SQL Analysis — Question 07

TRIP DISTANCE ANALYSIS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL Query

```sql
-- 7. Trip Distance Distribution: Determine the average trip_distance for each pickup_borough and
    dropoff_borough.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID, trip_distance
      FROM stg_uber_2017
    UNION ALL
    SELECT PULocationID, DOLocationID, trip_distance
      FROM stg_uber_2018
    UNION ALL
    SELECT PULocationID, DOLocationID, trip_distance
      FROM stg_uber_2019
    UNION ALL
    SELECT PULocationID, DOLocationID, trip_distance
      FROM stg_uber_2020
)
SELECT zp.Borough AS pickup_borough, zd.Borough AS dropoff_borough,
AVG(t.trip_distance) AS avg_distance FROM all_trips t
JOIN taxi_zones zp
    ON t.PULocationID = zp.LocationID
JOIN taxi_zones zd
    ON t.DOLocationID = zd.LocationID
GROUP BY
    zp.Borough,
    zd.Borough
ORDER BY avg_distance DESC;
```

Query Result: Trip Distance Distribution

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

pickup_borough	dropoff_borough	total_trips
Manhattan	Staten Island	96.16493506493514
Brooklyn	Bronx	31.581262626262617
Queens	Staten Island	30.522021505376355
Bronx	Staten Island	28.35581458145817
Unknown	Queens	26.180663615560654



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SQL Analysis — Question 08

PAYMENT METHOD ANALYSIS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL QUERY

```sql
#8. Payment Method Analysis: Calculate the percentage share of each payment_type and their average fare_amount.
WITH all_trips AS (
 SELECT payment_type, fare_amount
 FROM stg_uber_2017
 UNION ALL
 SELECT payment_type, fare_amount
 FROM stg_uber_2018
 UNION ALL
 SELECT payment_type, fare_amount FROM stg_uber_2019
 UNION ALL
 SELECT payment_type, fare_amount
 FROM stg_uber_2020
 )
 SELECT payment_type,
   COUNT(*) AS total_trips,ROUND( COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (),
   2
   ) AS percentage_share,
   ROUND(AVG(fare_amount), 2) AS avg_fare_amount
 FROM all_trips
 GROUP BY payment_type;
```

QUERY RESULT

payment_type	total_trips	percentage_share	avg_fare_amou
2	7174453	40.68	10.69
1	9397280	53.28	15.22
3	82897	0.47	3.01
4	36487	0.21	3.97
5	663	0.00	10.74

Result 8



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SQL Analysis — Question 09

YEARLY TRIP TRENDS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL Query

```sql
-- 9. High-Tip Routes: Find the top 5 pickup-dropoff combinations with the highest average tip_amount.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID, tip_amount
     FROM stg_uber_2017
    UNION ALL
    SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2018
    UNION ALL
    SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2019
    UNION ALL
    SELECT PULocationID, DOLocationID, tip_amount FROM stg_uber_2020
)
SELECT zp.Zone AS pickup_zone, zd.Zone AS dropoff_zone, COUNT(*) AS total_trips,
ROUND(AVG(t.tip_amount), 2) AS avg_tip
FROM all_trips t JOIN taxi_zones zp
 ON t.PULocationID = zp.LocationID
 JOIN taxi_zones zd ON t.DOLocationID = zd.LocationID
 WHERE t.tip_amount IS NOT NULL GROUP BY zp.Zone, zd.Zone
 HAVING COUNT(*) >= 10 ORDER BY avg_tip DESC
 LIMIT 5;
```

Query Result: Top 5 Pickup-Dropoff Combinations

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pickup_zone	dropoff_zone	total_trips	avg_tip
Washington Heights South	Heartland Village/Todt Hill	11	17.71
Morningside Heights	Eltingville/Annadale/Prince's Bay	11	17.71
LaGuardia Airport	DUMBO/Vinegar Hill	13	17.67
Flatbush/Ditmas Park	Newark Airport	10	17.51
Flushing	Newark Airport	10	17.39



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SQL Analysis — Question 10

YEARLY TRIP TRENDS

BETTER INSIGHTS
FOR A SMARTER RIDE

SQL Query

```sql
-- 10. Revenue by Borough Pairs: Calculate total_amount earned for each pickup_borough
-- to dropoff_borough pair.
WITH all_trips AS (
    SELECT PULocationID, DOLocationID, total_amount
    FROM stg_uber_2017
    UNION ALL
    SELECT PULocationID, DOLocationID, total_amount
    FROM stg_uber_2018
    UNION ALL
    SELECT PULocationID, DOLocationID, total_amount
    FROM stg_uber_2019
    UNION ALL
    SELECT PULocationID, DOLocationID, total_amount
    FROM stg_uber_2020
)
SELECT
  zp.Borough AS pickup_borough,
  zd.Borough AS dropoff_borough,
  COUNT(*) AS total_trips,ROUND(SUM(t.total_amount), 2) AS total_revenue
  FROM all_trips t JOIN taxi_zones zp ON t.PULocationID = zp.LocationID
  JOIN taxi_zones zd
ON t.DOLocationID = zd.LocationID
WHERE t.total_amount IS NOT NULL
GROUP BY
    zp.Borough,
    zd.Borough
ORDER BY total_revenue DESC;
```

Query Result: Revenue by Borough Pairs

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pickup_borough	dropoff_borough	total_trips	total_revenue
Manhattan	Manhattan	5220727	65094919.3
Brooklyn	Brooklyn	4273039	62748454.98
Queens	Queens	4465151	57802084.18
Brooklyn	Manhattan	774284	22919660.33
Bronx	Bronx	807010	12735130.67



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Trip And Revenue Overview

Trip And Revenue Overview

Time-Based Demand Analysis

Geographic Patterns

Payment & Tips Insights

Passenger & Distance Insights

month_name

All

trip_year

All



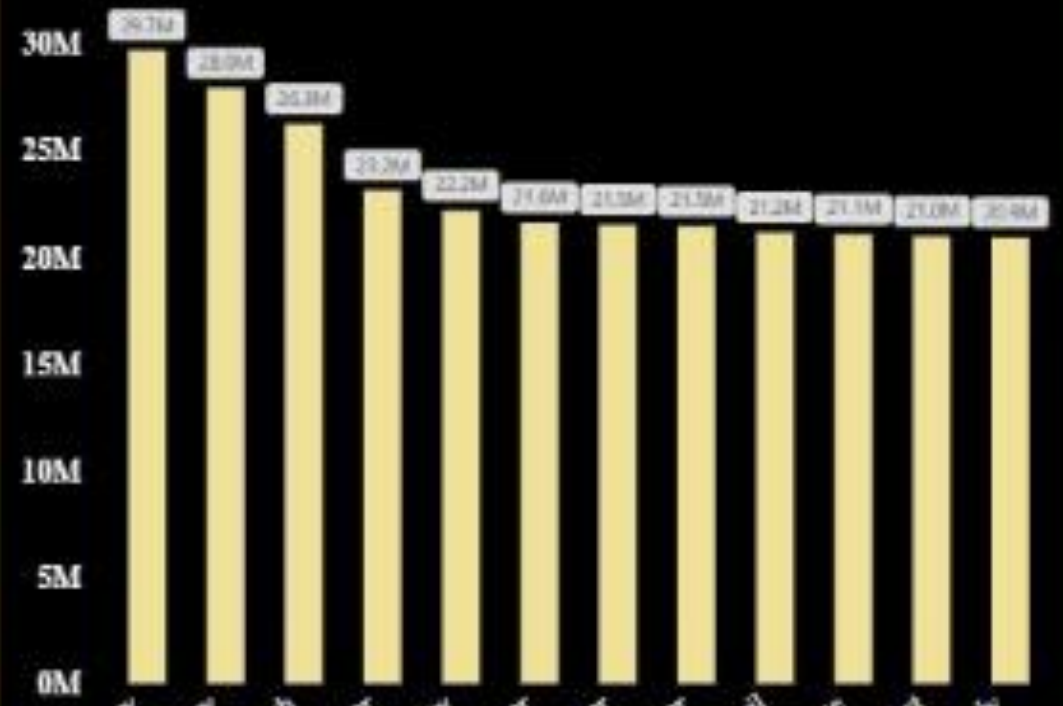
Total_Trip
18M

Total_Revenue
292.31M

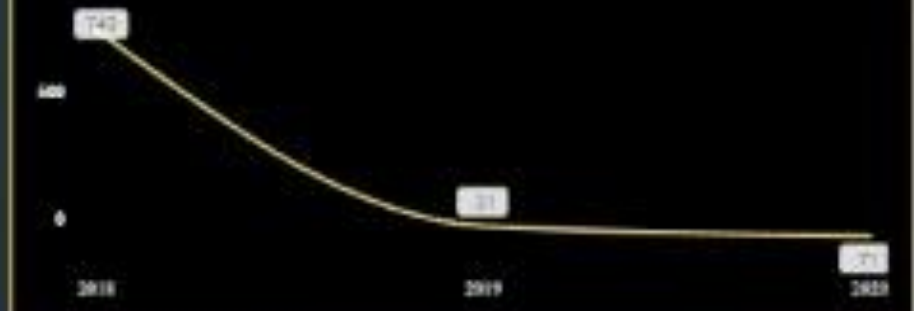
Avg_Fare_Trip
18.45

YOY_Trips
-4E-16

Monthly Revenue



Year over Year growth



Fare Distribution



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Time-Based Demand Analysis

- Trip And Revenue Overview
- Time-Based Demand Analysis
- Geographic Patterns
- Payment & Tips Insights
- Passenger & Distance Insights

Total_Trip
18M

Avg_Trip_Distance
4.58

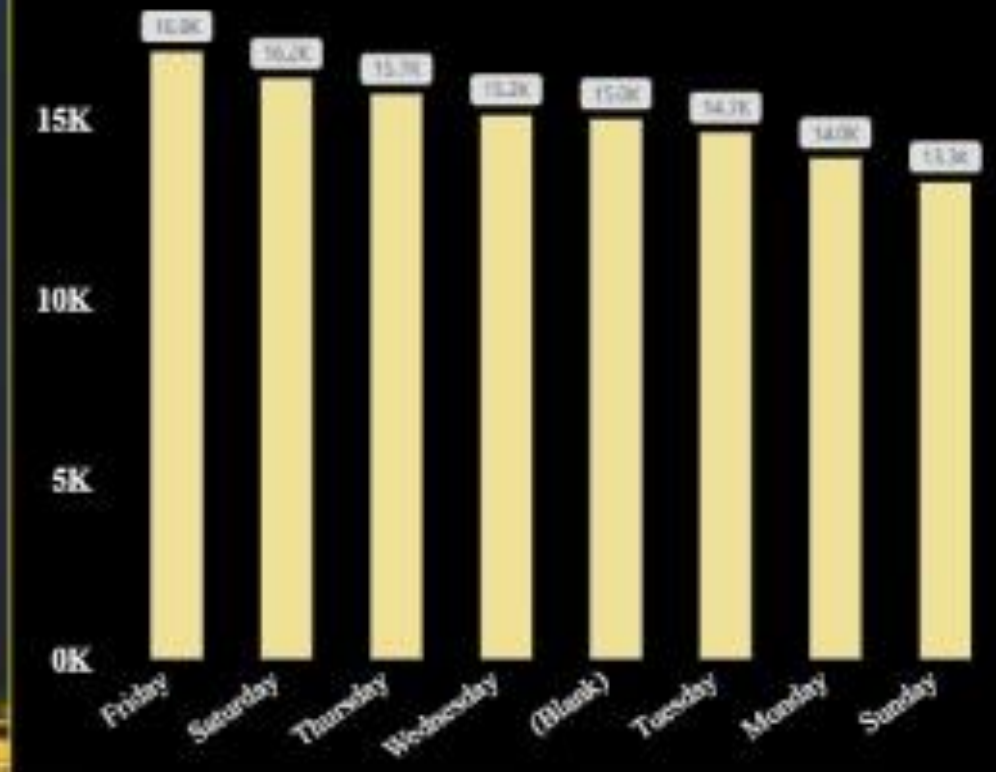
Avg_Fare_Trip
18.45

Total_Zone
262

trip_hour
All

day_of_week
All

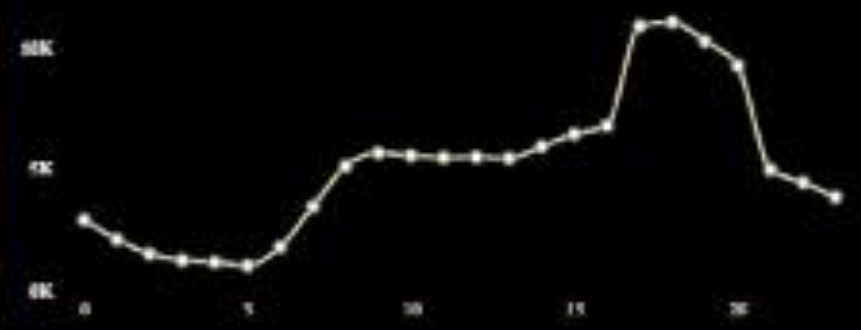
Trips by day



Fare Distribution



Trips by hour



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Geographic Patterns

- Trip And Revenue Overview
- Time-Based Demand Analysis -
- **Geographic Patterns**
- Payment & Tips Insights -
- Passenger & Distance Insights -

Zone: All

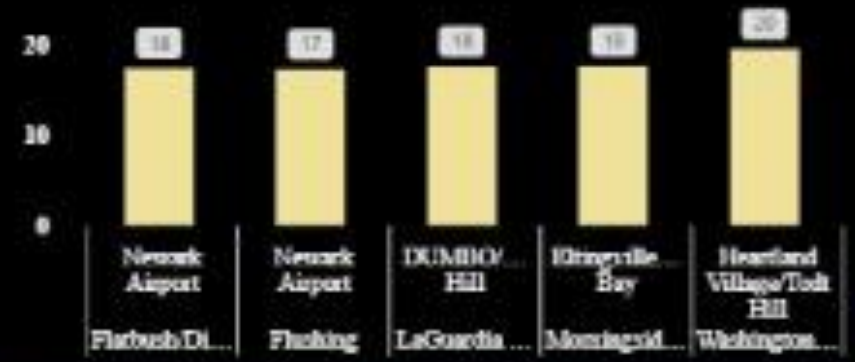
Borough: All

Total_Zone	Total_Borough	Avg_Trip_Distance	Total_Trip
262	7	4.58	18M

Trips by Borough



Avg trip by PickupZone and DropZone



Borough Analysis



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Passenger & Distance Insights

- Trip And Revenue Overview
- Time-Based Demand Analysis -
- Geographic Patterns
- Payment & Tips Insights -
- **Passenger & Distance Insights -**

trip_year
All

Borough
All



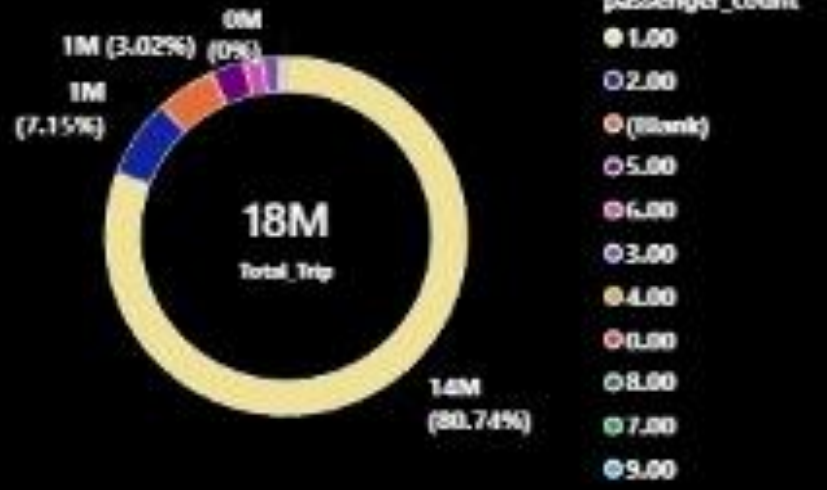


Avg_passenger_Trip
1.33

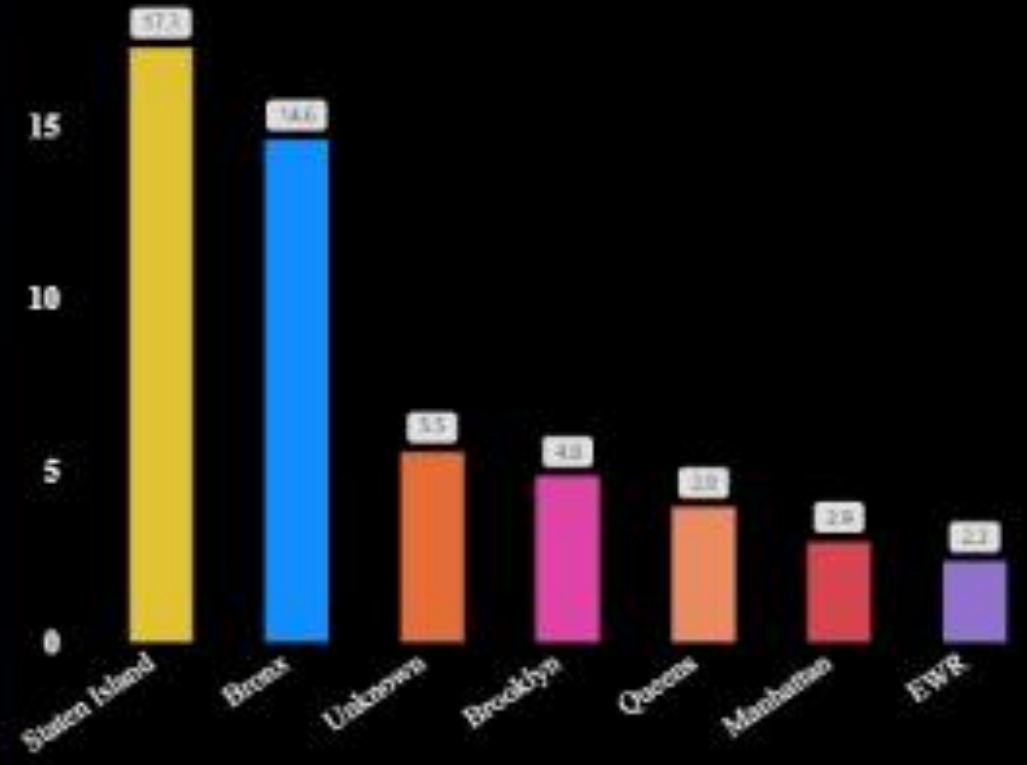


Avg_Trip_Distance
4.58

Total_trip by Passenger count



Avg_Trip Distance by Borough



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Payment & Tips Insights

- Trip And Revenue Overview
- Time-Based Demand Analysis -
- Geographic Patterns
- **Payment & Tips Insights -**
- Passenger & Distance Insights -

trip_year
All

Payment_Type_Label
All



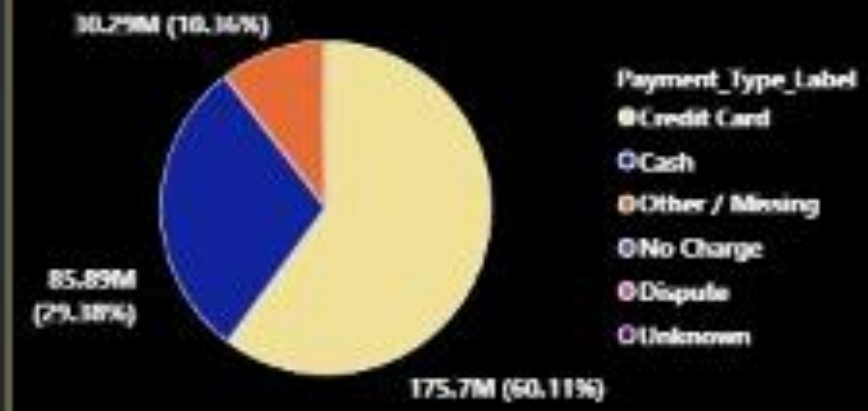


Avg_Tip
1.04

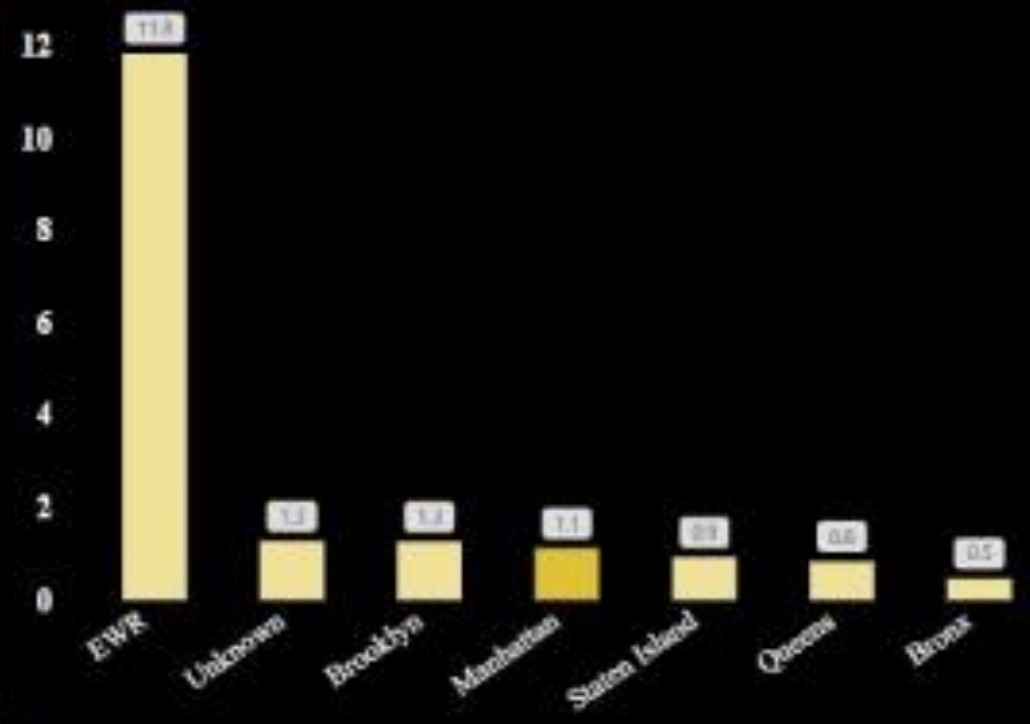


Payment_Type_Share
1.00

Payment Type Distribution



tip by borough



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Key Insights & Recommendations

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BETTER INSIGHTS
FOR A SMARTER RIDE

Fleet Scheduling

Key Insights (from SQL & Dashboard)

- Peak day and time analysis shows higher trip demand on specific days and time slots.
- Borough-wise trip distribution highlights areas with more ride activity.

Recommendations

- Allocate more drivers and vehicles during peak days and peak hours.
- Adjust fleet distribution across boroughs based on trip demand.

Pricing Optimization

Key Insights (from SQL & Dashboard)

- Monthly revenue trend shows variation across different months.
- Average revenue per trip provides understanding of earning pattern.
- Payment method analysis shows customer preferences.

Recommendations

- Review pricing strategy based on monthly revenue and trip value trends.
- Promote preferred payment methods to increase usage and revenue.

Passenger Service Improvement

Key Insights (from SQL & Dashboard)

- Route analysis shows the most used routes and popular trip patterns.
- Passenger and distance insights highlight travel behavior and trip characteristics.
- Tips analysis shows variation across payment methods and trip types.

Recommendations

- Focus on high-demand routes and improve service quality on those routes.
- Enhance customer experience for longer trips and preferred payment methods.



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Conclusion & Thank You

CAPSTONE PROJECT

BETTER INSIGHTS
FOR A SMARTER RIDE

Conclusion

- Analyzed Uber taxi trip data using SQL queries and Power BI dashboards to understand **trip trends, revenue patterns, passenger behavior and operational performance.**
- Identified **peak times, popular routes, borough-wise distribution, payment methods and tips patterns** through detailed analysis.
- Created **5 interactive Power BI dashboards** to provide a clear view of business performance and key trends.
- The insights help in **better decision-malking** for fleet management, pricing and passenger service improvements.

Thank You!

FOR YOUR TIME & SUPPORT

Prepared by
CHETAN AGRAWAL

